# FLEET TEAM DIGEST

## 737MAX-FTD-22-21002

**Issue Title :** SPEED TRIM FAIL Light Illumination Events Due to Single-Channel Trips of the Flight Control Computer (FCC) Trim Command Response Monitor (CRM)

| Airplane Model | ATA | Minor Model(s) |
|---|---|---|
| 737MAX | 2211-00 | -7, -8, -8200, -9, -BBJ |
| **Other Model(s)** | **ECCN** | |
| | 9E991 | |

| Originated Date | Last Revised Date | Created On |
|---|---|---|
| 04/30/2021 | 02/24/2026 | 02/24/2026 |
| **Estimated Completion Date** | **Next Update Date** | |
| 12/31/2026 | 09/30/2026 | |

## Revision Description

No changes were made to this Fleet Team Digest article.

## Applicability

All fielded 737 MAX aircraft operating with Flight Control Computer software version P12.1.2.

## Description

As part of 737 MAX Return to Service (RTS) activities, Boeing developed and certified the enhanced P12.1.2 Flight Control Computer (FCC) Operational Program Software (OPS). This new software incorporates enhancements to the stabilizer trim system, including new/modified monitors that control the SPEED TRIM FAIL light displayed within the flight deck.

Since late 2020, Boeing has been made aware of several in-service events that prompted illumination of the SPEED TRIM FAIL light on various MAX model aircraft. The vast majority of the SPEED TRIM FAIL reports have been driven by trips of the trim system's Command Response Monitor (CRM) within one of the aircraft's Flight Control Computers (FCC).

These faults are known as "single-channel" trips and will only be evident when the flight crew performs a Master Caution (MC) recall (usually initiated at the flight crew's top-of-descent checklist), which will cause the SPEED TRIM FAIL light to illuminate thereafter. After landing, maintenance personnel perform the Digital Flight Control System (DFCS) Built-In Test Equipment (BITE)

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checks per Aircraft Maintenance Manual (AMM) Pageblock 22-11-00/501 and typically find a fault logged (Maintenance Message 22-11186) during the climb phase within one of the aircraft's two FCC Fault History BITE Menus, as shown in the attached photo.

An in-service data review has determined that the above fault usually occurs after takeoff and climbing towards a cruise or intermediate level off altitude while the airplane is being operated with the autopilot off and the pilot is actively trimming the stabilizer using the main electric trim switches. Additionally, the fault has been observed to trigger shortly after flaps have been retracted, resulting in a "single-channel" trip of the speed trim command response monitor in the "command" FCC.

Should this scenario occur, control of the speed trim system will automatically transition from the "command" FCC to the "standby" FCC channel. Boeing also notes that the transition activity is not apparent to the flight crew during this time. Only later in the flight when a Master Caution recall is performed, will the FLT CONT system annunciator light illuminate on the Captain's side, prompting the flight crew to look up at the Flight Controls Panel, observing the SPEED TRIM FAIL light has now illuminated.

In all cases, the aircraft’s Speed Trim System (STS) continued to function normally without any need for flight crew action. The intended function is for the STS to become inhibited in the "command" FCC and STS control is passed to the "standby" FCC. After which, the STS inhibit is latched in the "command" FCC until maintenance action is taken or power is cycled. The "standby" FCC continues to control the STS and the CRM is then enabled in the second FCC.

At this time, Boeing considers these reports to be “nuisance” events with no detrimental airplane effects expected.

## Background

The trim CRM is an FCC internal stabilizer monitor that is active during both manual and autopilot flight. During manual flight, the CRM can cause the SPEED TRIM FAIL light to illuminate. During autopilot flight, the CRM can case the STAB OUT OF TRIM light to illuminate. Furthermore, the CRM also exists on the 737NG models and was continued on the 737 MAX with no changes to its operation. The primary functions of the CRM are to detect out-of-trim conditions manifested by either lack of stabilizer motion in response to an FCC-issued trim command or the prolonged duration of stabilizer trim movement. The P12.1.2 FCC OPS updates for the 737 MAX model did not change the CRM logic; however, it introduced a new logic level source that could cause the CRM to activate. This new source is the stabilizer trim voting algorithm that is performed by the FCCs to support the Cross-FCC Trim Monitor on 737 MAX models.

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As stated previously, the original CRM logic and fault responses are unchanged moving from the prior P11.1 FCC OPS to the P12.1.2 FCC OPS. Rather, the P12.1.2 FCC OPS introduced the stabilizer trim command voting function between the left and right FCCs such that the FCCs must both determine and agree that trim is to be commanded. This results in two trim calculations; one by the "command" FCC, and one by the "standby" FCC. This logic was introduced to support the Cross-FCC Trim Monitor.

The "command" (active) FCC may calculate a trim command that indicates the stabilizer should move, but the "standby" FCC may disagree due to computation timing and small differences between the local sensor/data sources associated with each FCC. This disagreement may result in the "command" FCC calculating that trim is needed, but not issuing said command to the stabilizer trim motor (due to the disagreement). This is also referred to as a "dead trim" situation whereby the "command" FCC had previously determined that its command to trim is set, but is not observing stabilizer surface position movement. This results in activation of the CRM due to the "standby" FCC trim command disagreement thereby, causing the SPEED TRIM FAIL light to illuminate during MC recall.

If the two FCCs do finally determine and agree on what trim command to issue, then the “dead trim” situation described above is no longer present and the CRM trip will clear. However, the latched Master Caution recall fault will remain until maintenance action is taken to clear the fault (i.e. successful completion of DFCS BITE, cycling power to the FCC(s), replacing discrepant components, etc.).

For additional information on the holistic set of changes implemented in the P12.1.2 FCC OPS, please see the "MCAS Maintainer Overview" provided in Multi-Operator Message (MOM) MOM-MOM-21-0051 and Boeing Service Letters 737-SL-22-073 & 737-SL-22-074.

## Status

Findings from the in-service investigations are being used to update the FCC functionality per the forthcoming P13.0 FCC OPS, which is bundled with the 737-10 and enhanced AOA (eAOA) certification programs. This update is expected to mitigate the nuisance condition once fielded.

## Interim Action

Perform the Interactive Fault Isolation Manual tasks (IFIM), as applicable to the observed Speed Trim Faults listed below:

- SPEED TRIM FAIL Light Comes On When MASTER CAUTION Light Switch Pushed: Perform 737 IFIM Task 22-11-00-810-D71
- DFCS BITE Maintenance Message 22-11186 (DFCS TRIM CMD RESPSE MON-FCCs detect stabilizer trim servo command/ response problem): Perform 737 IFIM Task 22-11-00-810-C61

## Final Action

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All 737 MAX Operators will be encouraged to update the FCC to P13.0 when it becomes available in order to mitigate the nuisance condition described in this Fleet Team Digest article.

## Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

Root Cause Established: August 2021 (Complete)

Solution Selected: September 2021 (Complete)

P13.0 Retrofit Service Bulletin Available: Will follow post 737-10 ammended type certification activity completion

## Reference Categories

| Reference Type | Reference Number | Revision Number | Reference Date |
|---|---|---|---|
| Service Letter (SL) | 737-SL-22-073 | | |
| Service Letter (SL) | 737-SL-22-074 | | |
| Multi-Operator Message (MOM) | MOM-MOM-21-0051-01B | | |
| Fault Isolation Manual (FIM) | IFIM Task 22-11-00-810-D71 | | |
| Fault Isolation Manual (FIM) | IFIM Task 22-11-00-810-C61 | | |

## Parts List

| Part Name | Boeing Part Number | Supplier Name | Supplier Number |
|---|---|---|---|
| P12.1.2 FCC OPS | 831-5854-223 | Collins Aerospace | 2274-COL-AC2-26 |

## Related Categories

- Dispatch Reliability Issues : Issues that cause revenue flight delays and cancellations
- Flight Operations : Flight Operations

## Attachments:

a) NuisanceProfile

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